



Iteration 3

$$m = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}, b = 0.9318$$

$$x = 0.01$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$X_m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

$$= (1)(-1.3316) + (3)(1.1808)$$

$$= 2.2108$$

$$X_m = \begin{bmatrix} 2.2108 \\ 6.4816 \end{bmatrix}$$

$$(4)(-1.3316) + (10)(1.1808)$$

$$= 6.4816$$

$$\hat{y} = X_m + b$$

$$= \begin{bmatrix} 2.2108 \\ 6.4816 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$= \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

STRENGTHENING JUSTICE DELIVERY SYSTEMS FOR
REGIONAL INTEGRATION AND ECONOMIC GROWTH



$$\begin{aligned}\text{Error} &= \hat{y} - y \\ &= \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}\end{aligned}$$

$$\bar{Y} = \frac{\sum X^T E}{n}$$

$$\begin{aligned}n &= 2, \bar{Y} = X^T E \\ &= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix} = (3)(-1.8574) + (10)(1.4134) \\ &= (1)(-18.574) + (4)(1.4134) \\ &= 8.5618 \\ &= 3.7962\end{aligned}$$

$$\bar{Y}_m = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$\begin{aligned}\bar{Y}_b &= \frac{\sum E}{n} \\ &= \frac{-1.8574 + 1.4134}{2} \\ &= -0.4440\end{aligned}$$

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$$M_{new} = m - \alpha \bar{X}_m$$

$$= \begin{bmatrix} -1.5316 \\ 1.1808 \end{bmatrix} - 0.01 \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} - \begin{bmatrix} 0.037962 \\ 0.085618 \end{bmatrix}$$

$$M_{new} = \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix}$$

$$b_{new} = b - \alpha \bar{X}_b$$

$$= 0.9318 - (0.01)(-0.444)$$

$$= 0.9318 - 0.00444$$

$$= 0.9382$$